Erdős–Kac theorem

- Proof by A.Grandville and K.Soudararajan -

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December 30, 2025

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Introduction

Theorem statement

The main theorem which we are going to proof is the following:

Theorem 1.1 (Erdos-Kac theorem 1940)

For $N \geq 1$ consider the set $\Omega_N = \{1, 2, 3, \cdots, N\}$. Let $w(n)$ be the number of distinct prime factors of n for any positive integer $n \geq 1$. Consider the random variable $X_N = \frac{w(n) - loglogN}{\sqrt{loglogN}}$. Then for any real numbers $a \leq b$ we have,

$$\lim_{N \to \infty} \mathbb{P}(a \leq X_N \leq b) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{x^2}{2}} dx$$

Vaguely speaking the random variable X_N converges in law to the standard normal distribution as $N \to \infty$. In this presentation I will present two proofs of this theorem for which we need to come up with few prerequisites which will be heavily used in the proof.

Prerequistes

Prerequsites

Theorem 2.1

For $N \geq 1$, let $\Omega_N = \{1, \cdots, N\}$ with the uniform probability distribution. Fix an integer $q \geq 1$, and denote by $\pi_q : \mathbb{Z} \to \mathbb{Z}/q\mathbb{Z}$ the reduction modulo q map.Let X_N be the random variables given by $X_N(n) = \pi_q(n)$ for $n \in \Omega_N$. For any function,

$$f : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$$

we have

$$|E(f(X_N)) - E(f)| \leq \frac{2}{N}||f||_1$$

where $||f||_1 = \sum_{a \in \mathbb{Z}/q\mathbb{Z}} |f(a)|$.

Proof.

Observe X_N takes values only the residues of q. By the defination of expectation,

$$E(f(X_N)) = \frac{1}{N} \sum_{1 \leq n \leq N} f(\pi_q(n))$$

For any $a \in \mathbb{Z}/q\mathbb{Z}$, say $f(\pi_q(a))$ repeats by m_a times whenever $n \equiv a \pmod{q}$. So

$$\implies \frac{N-1}{q} - 1 \leq m_a \leq \frac{N-1}{q} + 1$$

or,

$$\left| m_a - \frac{N}{q} \right| \leq 1 + \frac{1}{q}$$

$\square$

Proof.

$$E(f(X_N)) = \frac{1}{N} \sum_{1 \leq n \leq N} f(\pi_q(n)) = \frac{1}{N} \sum_{a \in Z/qZ} f(a).m_a$$

. and

$$E(f) = \frac{1}{q} \sum_{a \in Z/qZ} f(a)$$

.

$$\begin{aligned} |E(f(X_N)) - E(f)| &= \left| \sum_{a \in Z/qZ} f(a) \left(\frac{m_a}{N} - \frac{1}{q} \right) \right| \\ &\leq \frac{1}{N} \sum_{a \in Z/qZ} f(a) \left| m_a - \frac{N}{q} \right| \\ &\leq \sum_{a \in Z/qZ} f(a) \left(\frac{1 + \frac{1}{q}}{N} \right) \leq \frac{2}{N} ||f||_1 \end{aligned}$$

□

From this theorem we can see that,

$$\lim_{N\to\infty} |E(f(X_N)) - E(f)| = 0$$

.

- **Question:** Fix a positive integer $q \geq 1$. Choose a number n from the set $\Omega_N = \{1, 2, \cdots, N\}$. What is the probabilty that $n \equiv a \pmod{q}$?

Intuitively it seems to be $\frac{1}{q}$ which is true but this can be easily proved by the above theorem. Consider a characteristic function $f : \mathbb{Z}/q\mathbb{Z} \to 0, 1$, which gives 1 is $n \equiv a \pmod{q}$ otherwise 0.

$$E(f(X_N)) = \mathbb{P}(X_N \equiv a \pmod{q})$$

and $E(f) = \frac{1}{q}$ Thus, by the above theorem,

$$\lim_{N\to\infty} |E(f(X_N)) - E(f)| = 0$$

$$\implies \lim_{N\to\infty} \mathbb{P}(X_N \equiv a \pmod{q}) = \frac{1}{q}$$

A fun exerice

Consider the set $\Omega_N = \{1, 2, \cdots, N\}$. From this set chose a number say n. Show that

$$\lim_{N \to \infty} \mathbb{P}_N(n \text{ is square free}) = \frac{6}{\pi^2}$$

Hints:

- Use inclusion and exclusion
- Use the fact $\sum_{n=1}^{\infty} \frac{\mu(n)}{n^2} = \frac{6}{\pi^2}$

Moment generating function

Definition

The Moment generating function of a random variable X is denoted by $M_X(t)$ which is defined by

$$M_X(t) = E(e^{tX})$$

Here we will mainly concentrate on the moment generating function of normal distribution.
Let X be a random variable normally distributed with mean μ and variance σ^2.

$$M_X(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx}.e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx$$

By using some series of calculations we can get that

$$M_X(t) = e^{\mu t + \frac{(\sigma t)^2}{2}}$$

. Now lets see that why it is called the moment generating function.

$$M_X(t) = E(e^{tX}) = E\left(\sum_{n=0}^{\infty} \frac{X^n}{n!} t^n\right)$$

Assume that $M_X(t)$ is finite for all $-\infty < t < \infty$. Thus, the last expression of $M_X(t)$ is permissible to interchange the order of expectation and summation. In other words,

$$M_X(t) = \sum_{n=0}^{\infty} \frac{E(X^n)}{n!} t^n$$

We are interested in the moments of Normal distributed random variable say X of mean 0 and variance σ^2. Thus lets find out the $M_X(t)$ of X.
We have,

$$M_X(t) = e^{\frac{(\sigma t)^2}{2}}$$

$$= \sum_{n=0}^{\infty} \frac{(\sigma t)^{2n}}{2^n} \frac{1}{n!}$$

Comparing the coefficients of t^k we get that the odd moments of X is 0, and

$$\frac{E(X^{2n})}{(2n)!} = \frac{\sigma^{2n}}{2^n.n!}$$

$$\implies E(X^{2n}) = \frac{(2n)!\sigma^{2n}}{2^n.n!}$$

for all $n \geq 0$.

Carleman's Condition

Condition

Let μ be a measure on $\mathbb{R}$ such that all the moments,

$$m_n = \int_{-\infty}^{\infty} x^n . d\mu(x)$$

are finite for all $n \geq 0$. If

$$\sum_{n=1}^{\infty} m_{2n}^{-\frac{1}{2n}} = +\infty$$

then μ is the only measure on $\mathbb{R}$ with m_n as its sequence of moments.

Carleman's condition gives a sufficient condition for the determinacy of the moment problem. That is, if a measure μ satisfies Carleman's condition, there is no other measure ν having the same moments that of μ.

Observe that gaussian distribution satisfies the Carleman's Condition.

Corrollary 1.1

Let X_N be a sequence of random variables satisfying the following:

- $$\lim_{N\to\infty} E(X_N^{2k+1}) = 0$$

  for all $k \geq 0$

- $$\lim_{N\to\infty} E(X_N^{2k}) = \frac{(2k)!\sigma^{2k}}{2^k k!}$$

  for all $k \geq 0$

Say, $\lim_{N\to\infty} X_N = Y$. Then $Y \sim N(0, \sigma^2)$.

Mertens Theorem

Theorem

For any real number $x \geq 1$,

$$\sum_{p \leq x} \frac{1}{p} = \ln(\ln x) + M + \mathcal{O}\left(\frac{1}{\ln x}\right)$$

Or,

$$\sum_{p \leq x} \frac{1}{p} \sim \ln(\ln x)$$

as $x \to \infty$.

One can get the proof of this theorem from internet source or from the Apostol book of The Introduction on Analytic Number Theory.

Lyapunov CLT

Theorem 2.2

Suppose $\{X_i\}_{i\geq 1}$ *be sequence of independent random variables with finite expectation* μ_i *and variance* σ_i^2*. Define* $s_n^2 = \sum_{i=1}^n \sigma_i^2$*.If for some* $\delta > 0$*, the Lyapunov condition:*

$$\lim_{n\to\infty} \frac{1}{s_n^{2+\delta}} \sum_{i=1}^{n} E[|X_i - \mu_i|^{2+\delta}] = 0$$

. satisfies. Then,

$$\lim_{n\to\infty} \frac{1}{s_n} \sum_{i=1}^{n} (X_i - \mu_i) \sim N(0,1)$$

Gcd restricted sum

theorem

Let $r = \prod_i^s p_i^{\alpha_i}$ and define $R = \prod_i^s p_i$. And say $\gcd(n, R) = d$. Then,

$$\sum_{\substack{n \leq x \\ (n,R)=d}} 1 = \left(\frac{x}{R}\right) \phi\left(\frac{R}{d}\right) + \mathcal{O}\left(\tau\left(\frac{R}{d}\right)\right)$$

Proof:

As we need to count n when $(n, R) = d$, we can say $n = md$. Thus,

$$\begin{aligned} \sum_{\substack{n \leq x \\ (n,R)=d}} 1 &= \sum_{\substack{md \leq x \\ (md,R)=d}} 1 \\ &= \sum_{\substack{m \leq \frac{x}{d} \\ (m,\frac{R}{d})=1}} 1 \end{aligned}$$

We can rewrite our sum in the following way,

$$\begin{aligned}
\sum_{\substack{n\leq x\\(n,R)=d}} 1 = \sum_{\substack{m\leq \frac{x}{d}\\(m,\frac{R}{d})=1}} 1 &= \sum_{m\leq \frac{x}{d}} \sum_{k|(m,\frac{R}{d})} \mu(k) \\
&= \sum_{k|(m,\frac{R}{d})} \mu(k) \sum_{\substack{m\leq \frac{x}{d}\\k|m}} 1. \\
&= \sum_{k|(m,\frac{R}{d})} \mu(k) \left(\frac{x}{dk} + \mathcal{O}(1)\right) \\
&= \frac{x}{d} \sum_{k|(m,\frac{R}{d})} \frac{\mu(k)}{k} + \sum_{k|(m,\frac{R}{d})} \mathcal{O}(1) \\
&= \frac{x}{R}\phi\left(\frac{R}{d}\right) + \mathcal{O}\left(\tau\left(\frac{R}{d}\right)\right).
\end{aligned}$$

Heuristic proof

An Heuristic proof of Erdos-Kacs Theorem

Consider the bernolli random variable X_p which gives 1 if $p|n$ else gives 0 for each prime p. Observe that we can write

$$w(n) = \sum_{p \leq n} X_p(n)$$

Since X_p is a bernolli RV, $E(X_p) = P(X_p = 1) = \frac{1}{p}$ and $\sigma_p^2 = \frac{1}{p}\left(1 - \frac{1}{p}\right)$. Thus,

$$\sum_p E(X_p) = \sum_p \frac{1}{p}$$

and

$$\sum_p \sigma_p^2 = \sum_p \frac{1}{p}\left(1 - \frac{1}{p}\right)$$

Here X_p are independent random variables satisfyig the Lyapunov's condition. Thus, using the theorem we can say,

$$\lim_{n\to\infty} \frac{(\sum_{p\leq n} X_p) - (\sum_{p\leq n} \mu_i)}{\sqrt{\sum_{p\leq n} \sigma_p^2}} \sim N(0,1)$$

As $n \to \infty$, $\sum_{p\leq n} X_p = w(n)$. By using the Mertens theorem, We can say that,

$$\lim_{n\to\infty} \frac{w(n) - \ln(\ln n)}{\sqrt{\ln(\ln n)}} \sim N(0,1)$$

This is how we come up with a non rigorous proof of Erdos kacs theorem.

The Proof by Granville and Soundararajan

PART I

Let p be a prime and $n \in \mathbb{N}$. Define $f_p(n)$ by,

$$f_p(n) = \begin{cases} 1 - \frac{1}{p} & \text{if } p \mid n \\ -\frac{1}{p} & \text{if } p \nmid n. \end{cases}$$ Denote,

$$C_k = \frac{\Gamma(k+1)}{2^{\frac{k}{2}}\Gamma(\frac{k}{2}+1)}$$

for every $k \in \mathbb{N}$. By using the property

$$\Gamma(k+1) = k\Gamma(k)$$

. One can show that

$$C_k = \frac{k!}{2^{k/2}\left(\frac{k}{2}\right)!}$$

Proposition

Let $z \geq 10^6$ be a real number. For even natural numbers $k \leq (\log\log z)^{\frac{1}{3}}$,

$$\sum_{n \leq x} \left(\sum_{p \leq z} f_p(n) \right)^k = C_k x (\log\log z)^{\frac{k}{2}} \left(1 + \mathcal{O}\left(\frac{k^3}{\log\log z} \right) \right) + \mathcal{O}(3^k . \pi(z)^k).$$

And for odd natural numbers k,

$$\sum_{n \leq x} \left(\sum_{p \leq z} f_p(n) \right)^k \ll C_k x (\log\log z)^{\frac{k}{2}} \frac{k^{\frac{3}{2}}}{\sqrt{\log\log z}} + \mathcal{O}(3^k \pi(z)^k).$$

Proof of Proposition

Observe that $f_p(n)$ is multiplicative. Thus for $n = \prod_i p_i^{\alpha_i}$, we have

$$f_r(n) = \prod_i f_{p_i}(n)^{\alpha_i}$$

$$\sum_{n \le x} \left(\sum_{p \le z} f_p(n) \right)^k = \sum_{p_1, p_2, \cdots, p_k \le z} \sum_{n \le x} f_{p_1 \cdots p_k}(n)$$

Now lets change our focus on finding some cool stuffs on $\sum_{n \le x} f_r(n)$. For $r = \prod_i^s q^{a_i}$, let $R = q_1 \cdot q_2 \cdots q_s$. Observe that,

$$f_r(n) = f_r(\gcd(n, R))$$

Thus we can write,

$$\sum_{n\le x} f_r(n) = \sum_{d|R} f_r(d) \sum_{\substack{n\le x \\ (n,R)=d}} 1.$$

$$= \sum_{d|R} f_r(d) \left[\frac{x}{R}\phi\left(\frac{R}{d}\right) + \mathcal{O}\left(\tau\left(\frac{R}{d}\right)\right)\right]$$

$$= \frac{x}{R}\sum_{d|R} f_r(d)\phi\left(\frac{R}{d}\right) + \mathcal{O}\left(\sum_{d|R} f_r(d)\tau\left(\frac{R}{d}\right)\right).$$

Denote $G(r) = \frac{1}{R}\sum_{d|R} f_r(d)\phi\left(\frac{R}{d}\right)$. So we get,

$$\sum_{n\le x} f_r(n) = xG(r) + \mathcal{O}\left(\sum_{d|R}\tau\left(\frac{R}{d}\right)\right)$$

By combinatorial arguments one can show that,

$$\sum_{d|R}\tau\left(\frac{R}{d}\right) = \sum_{i=0}\binom{s}{i}2^{s-i} = 3^s$$

Thus,

$$\sum_{n \leq x} f_r(n) = xG(r) + \mathcal{O}(3^s)$$

Observe that $G(r)$ is convolution, since f_r and ϕ are multiplicative function we can say that $G(r)$ is also multiplicative. For $r = p^\alpha$, $R = p$.

$$\begin{aligned} G(r) &= \frac{1}{p} \sum_{d|p} f_r(d) \phi\left(\frac{p}{d}\right) \\ &= \frac{1}{p} \left(f_r(1)\phi(p) + f_r(p)\phi(1)\right) = \frac{1}{p}\left[\left(-\frac{1}{p}\right)^\alpha (p-1) + \left(1 - \frac{1}{p}\right)^\alpha\right] \end{aligned}$$

If $\alpha = 1$ for any prime factor of r, then $G(r) = 0$.

So,

$$\begin{aligned}
\sum_{n\le x}\left(\sum_{p\le z} f_p(n)\right)^k &= \sum_{p_1,p_2,\cdots,p_k\le z}\ \sum_{n\le x} f_{p_1\cdots p_k}(n)\\
&= \sum_{p_1,\cdots,p_k\le z} xG(p_1\cdot p_2\cdots p_k)+\mathcal{O}(3^k)\\
&= \sum_{p_1,\cdots,p_k\le z} xG(p_1\cdot p_2\cdots p_k)+\sum_{p_1,\cdots,p_k\le z}\mathcal{O}(3^k)\\
&= \sum_{p_1,\cdots,p_k\le z} xG(p_1\cdot p_2\cdots p_k)+\mathcal{O}(3^k\pi(z)^k)
\end{aligned}$$

Among the p_i primes which are not necessarily unique, say $q_1 < q_2 < \cdots < q_s$ be the set of distinct primes. Also $s \le \frac{k}{2}$ since $v_{q_i}(r) \ge 2$.

We can update our result by,

$$\sum_{p_1.p_2.\cdots p_k \leq z} G(p_1.p_2.\cdots p_k) = \sum_{s \leq \frac{k}{2}} \sum_{q_1 < q_2 < \cdots < q_s} \sum_{\substack{\alpha_i \geq 2 \\ \sum_i \alpha_i = k}} \frac{k!}{\alpha_i! \cdots \alpha_s!} G(q_1^{\alpha_1} \cdots q_s^{\alpha_s})$$

For k even, $s = \frac{k}{2}$ many distinct prime factors contribute in the above sum. Firstly just focus on the term when $s = \frac{k}{2}$ and $\alpha_i = 2$.

$$\sum_{q_1 < q_2 < \cdots < q_{\frac{k}{2}}} \frac{k!}{2^{\frac{k}{2}}} G(q_1^2.q_2^2 \cdots q_{\frac{k}{2}}^2) = \frac{k!}{2^{\frac{k}{2}} \left(\frac{k}{2}\right)!} \sum_{q_1, q_2, \cdots, q_{\frac{k}{2}} \leq z} \prod_i^{\frac{k}{2}} \frac{1}{q_i} \left(1 - \frac{1}{q_i}\right)$$

Now lets bound the above sum.

Exercise

Show that

$$\left(\sum_{\pi_{\frac{k}{2}} \le p \le z} \frac{1}{p}\left(1-\frac{1}{p}\right)\right)^{\frac{k}{2}} \le \sum_{q_1, q_2, \cdots, q_{\frac{k}{2}} \le z} \prod_i^{\frac{k}{2}} \frac{1}{q_i}\left(1-\frac{1}{q_i}\right) \le \left(\sum_{p \le z} \frac{1}{p}\left(1-\frac{1}{p}\right)\right)^{\frac{k}{2}}$$

Hint: $\frac{1}{p}\left(1-\frac{1}{p}\right)$ decreases as p increases.

Using the above result and applying the Mertens Theorem, we get

$$\begin{aligned} C_k \sum_{q_1, q_2, \cdots, q_{\frac{k}{2}} \le z} \prod_i^{\frac{k}{2}} \frac{1}{q_i}\left(1-\frac{1}{q_i}\right) &= \left(\sum_{p \le z} \frac{1}{p} - \sum_{p \le z} \frac{1}{p^2} + \mathcal{O}(\log\log k)\right)^{\frac{k}{2}} \\ &= (\log\log z + \mathcal{O}(1 + \log\log k))^{\frac{k}{2}}. \end{aligned}$$

$$(\log\log z + \mathcal{O}(1+\log\log k))^{\frac{k}{2}} = (\log\log z)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\frac{\log\log k}{\log\log z}\right)\right)^{\frac{k}{2}}$$
$$= (\log\log z)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\sum_{i=1}\binom{\frac{k}{2}}{i}\left(\frac{\log\log k}{\log\log z}\right)^{i}\right)\right)$$

All the terms of the inside sum is less than the first term $\frac{k}{2}\frac{\log\log k}{\log\log z}$.

$$(\log\log z + \mathcal{O}(1+\log\log k))^{\frac{k}{2}} = (\log\log z)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\frac{k}{2}\frac{k\cdot\log\log k}{\log\log z}\right)\right)$$
$$= (\log\log z)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\frac{k^3}{\log\log z}\right)\right)$$

Note: This is only the term when $s = \frac{k}{2}$. Now we will deal with the terms $s < \frac{k}{2}$.

We have

$$G(q_1^{\alpha_1} \cdot q_2^{\alpha_2} \cdots q_s^{\alpha_s}) = \prod_i^s \left[\left(-\frac{1}{q}\right)^{\alpha_i} \left(1 - \frac{1}{q}\right) + \frac{1}{q} \left(1 - \frac{1}{q}\right)^{\alpha_i} \right]$$

As α increases,

$$\left(-\frac{1}{q}\right)^{\alpha} \left(1 - \frac{1}{q}\right) + \frac{1}{q} \left(1 - \frac{1}{q}\right)^{\alpha}$$

decreases. Thus for $\alpha \geq 2$, it is maximum for $\alpha = 2$. Thus,

$$0 \leq \left(-\frac{1}{q}\right)^{\alpha} \left(1 - \frac{1}{q}\right) + \frac{1}{q} \left(1 - \frac{1}{q}\right)^{\alpha} \Big|_{\alpha=2} \leq \frac{1}{q}$$

Hence we can say,

$$0 \leq G(r) = G(q_1^{\alpha_1} \cdot q_2^{\alpha_2} \cdots q_s^{\alpha_s}) \leq \frac{1}{q_1 \cdot q_2 \cdots q_s}$$

Denote,

$$A = \sum_{s < \frac{k}{2}} \sum_{q_1 < q_2 < \cdots < q_s \leq z} \sum_{\substack{\alpha_i \geq 2 \\ \sum_i \alpha_i = k}} \frac{k!}{\alpha_1! \cdot \alpha_2! \cdots \alpha_s!} G(q_1^{\alpha_1} \cdot q_2^{\alpha_2} \cdots q_s^{\alpha_s})$$

So,

$$A \leq \sum_{s<\frac{k}{2}} \sum_{q_1<q_2<\cdots<q_s} \frac{k!}{q_1 \cdot q_2 \cdots q_s} \sum_{\substack{\alpha_i \geq 2 \\ \sum_i \alpha_i = k}} \frac{1}{\alpha_1! \cdot \alpha_2! \cdots \alpha_s!}$$

Alike the bound used in the last exercise and since $\alpha_i \geq 2$ we can show that,

$$\begin{aligned} A &\leq \sum_{s<\frac{k}{2}} \frac{k!}{s!} \left(\sum_{q \leq z} \frac{1}{q} \right)^s \sum_{\substack{\alpha_i \geq 2 \\ \sum_i \alpha_i = k}} \frac{1}{\alpha_1! \cdot \alpha_2! \cdots \alpha_s!} \\ &\leq \sum_{s<\frac{k}{2}} \frac{k!}{s! \cdot 2^s} \left(\sum_{q \leq z} \frac{1}{q} \right)^s \mathcal{H}_k(s) \end{aligned}$$

where $\mathcal{H}_k(s)$ denotes the number of s tuples $(\alpha_1, \alpha_2, \cdots, \alpha_s)$ such that $\alpha_i \geq 2$ and $\sum_i \alpha_i = k$.

$$\sum_i^s \alpha_i = k$$

$$\sum_i^s (\alpha_i - 2) = k - 2s$$

Thus,

$$\mathcal{H}_k(s) = \binom{k-s-1}{s-1}$$

Hence by using the Merten's Theorem we get,

$$A \leq \sum_{s<\frac{k}{2}} \frac{k!}{2^s \cdot s!} \binom{k-s-1}{s-1} (\log\log z + \mathcal{O}(1))^s$$

Case 1: k is even.
Using the fact $k^3 \leq \log\log z$ and the definition of A, C_k we get,

$$A \leq \sum_{s \leq \frac{k}{2}-1} \frac{k!}{2^s s!} \binom{k-s-1}{s-1} (\log\log z + \mathcal{O}(1))^s$$

$$\ll C_k (\log\log z)^{(\frac{k}{2}-1)} \sum_{s \leq \frac{k}{2}-1} \frac{k!}{2^s s! C_k} \binom{k-s}{s} \left(\frac{1}{\log\log z}\right)^{\frac{k}{2}-1-s}$$

$$\ll C_k \frac{(\log\log z)^{\frac{k}{2}}}{\log\log z} \sum_{s \leq \frac{k}{2}-1} \frac{k!}{2^s s! C_k} \binom{k-s}{s} \left(\frac{1}{k^3}\right)^{\frac{k}{2}-1-s}$$

$$\ll C_k (\log\log z)^{\frac{k}{2}} \left(\frac{k^3}{\log\log z}\right) \left(\sum_{s \leq \frac{k}{2}-1} \frac{k!}{2^s s! C_k} \binom{k-s}{s} \left(\frac{1}{k^3}\right)^{\frac{k}{2}-s}\right)$$

Exercise

Show that :

$$\sum_{s\leq \frac{k}{2}-1} \frac{k!}{2^s s! C_k} \binom{k-s}{s} \left(\frac{1}{k^3}\right)^{\frac{k}{2}-s} \ll 1$$

From the above exercise we can conclude that

$$A = C_k (\log\log z)^{\frac{k}{2}} \mathcal{O}\left(\frac{k^3}{\log\log z}\right)$$

So we are on the final part of the proof.

$$\begin{aligned}
\sum_{n\leq x}\left(\sum_{p\leq z} f_p(n)\right)^k &= x\left(\sum_{p_1,p_2,\cdots,p_k\leq z} G(p_1\cdot p_2\cdots p_k)\right) + \mathcal{O}(3^k\pi(z)^k) \\
&= x\left(\sum_{s\leq\frac{k}{2}}\sum_{q_1<q_2<\cdots<q_s\leq z}\sum_{\substack{\alpha_i\geq 2\\ \sum_i \alpha_i = k}} \frac{k!}{\alpha_1!\cdot\alpha_2!\cdots\alpha_s!} G(q_1^{\alpha_1}\cdot q_2^{\alpha_2}\cdots q_s^{\alpha_s})\right) \\
&+ \mathcal{O}(3^k\pi(z)^k)
\end{aligned}$$

$$\sum_{n\le x}\left(\sum_{p\le z} f_p(n)\right)^k = x\left(C_k\cdot(\log\log z)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\frac{k^3}{\log\log z}\right)\right)+A\right)+\mathcal{O}(3^k\pi(z)$$

$$= xC_k(\log\log z)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\frac{k^3}{\log\log z}\right)\right)+\mathcal{O}(3^k\pi(z)^k)$$

This completes the proof of the first proposition which is for k even. **The other case for odd k is left as an exercise for the readers.**

This is the proof of moments given by Granville and Soundarajan. For $N \geq 1$ consider the sequence of random variables,

$$X_N = \frac{w(n) - \log\log N}{\sqrt{\log\log N}}$$

for all $n \leq N$. If we can show that $\lim_{N\to\infty} E(X_N^k) = E(Y^k)$ where $Y \sim N(0,1)$ for all $k \geq 1$, then from (corollary 1.1) we can conclude that

$$\lim_{N\to\infty} X_N \sim N(0,1)$$

which completes the proof. For any $k \geq 1$,

$$E(X_N^k) = \frac{1}{N}\sum_{n\leq N}\left(\frac{w(n) - \log\log N}{\sqrt{\log\log N}}\right)^k$$

Claim

For even $k \in \mathbb{N}$ and $k \leq (\log\log N)^{\frac{1}{3}}$,

$$\sum_{n \leq N} (w(n) - \log\log N)^k = NC_k(\log\log N)^{\frac{k}{2}} \left(1 + \mathcal{O}\left(\frac{k^3}{\log\log N}\right)\right)$$

and for odd $k \in \mathbb{N}$ we have,

$$\sum_{n \leq N} (w(n) - \log\log N)^k \ll NC_k(\log\log N)^{\frac{k}{2}} \frac{k^{\frac{3}{2}}}{\sqrt{\log\log N}}$$

Proving this claim finishes our proof as by taking $N \to \infty$ we will obtain our required target. One can find similarities with the previous proposition. The proof of this claim is followed in the next part.

Final Part

Set $z = N^{\frac{1}{k}}$ and by the definition of $f_p(n)$ in the previous parts we get,

$$\left(f_p(n) + \frac{1}{p}\right) = \begin{cases} 1 & \text{if } p \mid n \\ 0 & \text{if } p \nmid n. \end{cases}$$

$$w(n) = \sum_{p \leq z} \left(f_p(n) + \frac{1}{p}\right) + \sum_{\substack{p > z \\ p \mid n}} 1.$$

There are atmost k many prime factors p of n such that $p > z > n^{\frac{1}{k}}$. Thus,

$$\begin{aligned} w(n) - \log\log N &= \sum_{p \leq z} f_p(n) + \mathcal{O}(k) + \log\log z - \log\log N + \mathcal{O}(1). \\ &= \left(\sum_{p \leq z} f_p(n)\right) + \mathcal{O}(k) - \log k = \left(\sum_{p \leq z} f_p(n)\right) + \mathcal{O}(k). \end{aligned}$$

$$
\begin{aligned}
(w(n) - \log\log N)^k &= \left(\sum_{p\le z} f_p(n) + \mathcal{O}(k)\right)^k \\
&= \left(\sum_{p\le z} f_p(n)\right)^k + \sum_{i=0}^{k-1} \binom{k}{i} \left(\sum_{p\le z} f_p(n)\right)^i \mathcal{O}(k)^{k-i} \\
&= \left(\sum_{p\le z} f_p(n)\right)^k + \mathcal{O}\left((ck)^{k-l}\binom{k}{l}\left|\sum_{p\le z} f_p(n)\right|^l\right)
\end{aligned}
$$

where l is the number for which the term $\binom{k}{l}\left(\sum_{p\le z} f_p(n)\right)^l \mathcal{O}(k)^{k-l}$ is maximum.

Summing over $n \leq N$ we get,

$$\sum_{n \leq N} (w(n) + \log\log N)^k = \sum_{n \leq N} \left(\sum_{p \leq z} f_p(n) \right)^k + \mathcal{O}\left(\sum_{n \leq N} (ck)^{k-l} \binom{k}{l} \left| \sum_{p \leq z} f_p(n) \right|^l \right)$$

For different n we will get different value of l in the above equation, thus $l = f(n)$. Consider the sets $A_l = \{n : f(n) = l \text{ and } n \leq N\}$. Here $l \leq k - 1$. If l is even then it can be directly handled by the proposition. If l is odd,

$$\sum_{n \in A_l} \left(\sum_{p \leq z} f_p(n) \right)^l \leq \sum_{n \leq N} \left(\sum_{p \leq z} f_p(n) \right)^l$$

By using Cauchy Swartz Inequality we get,

$$\sum_{n \leq N} \left(\sum_{p \leq z} f_p(n) \right)^l \leq \left(\sqrt{\sum_{n \leq N} \left(\sum_{p \leq z} f_p(n) \right)^{l+1}} \right) \left(\sqrt{\sum_{n \leq N} \left(\sum_{p \leq z} f_p(n) \right)^{l-1}} \right)$$

Now, applying the proposition on the above terms for even $l-1$ and $l+1$ we get,

$$\sum_{n\leq N}\left(\sum_{p\leq z} f_p(n)\right)^l \leq \left(NC_{l+1}(\log\log z)^{\frac{l+1}{2}}\left(1+\mathcal{O}\left(\frac{(l+1)^3}{\log\log z}\right)\right)+\mathcal{O}(3^{l+1}\pi(z)^{l+1})\right)^{\frac{1}{2}}\cdot$$

$$\left(NC_{l-1}(\log\log z)^{\frac{l-1}{2}}\left(1+\mathcal{O}\left(\frac{(l-1)^3}{\log\log z}\right)\right)+\mathcal{O}(3^{l-1}\pi(z)^{l-1})\right)^{\frac{1}{2}}\cdots(*)$$

Using the fact that $z=N^{\frac{1}{k}}$, notice

$$\begin{aligned}3^{l+1}\pi(z)^{l+1}\leq 3^k\pi(z)^k &\leq 3^k\left(\frac{z}{\log z}\right)^k\\ &\leq 3^k\left(\frac{Nk^k}{(\log N)^k}\right) << N\end{aligned}$$

Thus,

$$\mathcal{O}(3^{l+1}\pi(z)^{l+1}) << N$$

which is very negligible in comparison to the other terms.

The main dominating terms in $(*)$ are the $NC_{l+1}(\log\log z)^{\frac{l+1}{2}}$ and $NC_{l-1}(\log\log z)^{\frac{l-1}{2}}$. Thus,

$$\begin{aligned}\sum_{n\le N}\left(\sum_{p\le z} f_p(n)\right)^l &\ll \left(NC_{l+1}(\log\log z)^{\frac{l+1}{2}}\right)^{\frac{1}{2}}\left(NC_{l-1}(\log\log z)^{\frac{l-1}{2}}\right)^{\frac{1}{2}}\\ &\ll N\sqrt{C_{l-1}C_{l+1}}(\log\log z)^{\frac{l}{2}}\end{aligned}$$

So,

$$\begin{aligned}\sum_{n\le N}(w(n)-\log\log N)^k-\sum_{n\le N}\left(\sum_{p\le z} f_p(n)\right)^k &= \mathcal{O}\left(\sum_{\substack{l=0\\ 2|l}}^{k-1}(ck)^{k-l}\binom{k}{l}\sum_{n\in A_l}\left|\sum_{p\le z} f_p(n)\right|^l\right)\\ &\quad+\mathcal{O}\left(\sum_{\substack{l=0\\ 2\nmid l}}^{k-1}(ck)^{k-l}\binom{k}{l}\sum_{n\in A_l}\left|\sum_{p\le z} f_p(n)\right|^l\right)\end{aligned}$$

Lets focus on the right hand side of the inequality.Using the above bounds we get,

$$\sum_{n\leq N}(w(n)-\log\log N)^k-\sum_{n\leq N}\left(\sum_{p\leq z}f_p(n)\right)^k=\mathcal{O}\left(\sum_{\substack{l=0\\2|l}}^{k-1}(ck)^{k-l}\binom{k}{l}\left[C_lN(\log\log z)^{\frac{l}{2}}\left(1+\mathcal{O}\left(\frac{l^3}{\log\log z}\right)\right)\right]\right)$$
$$+\mathcal{O}\left(\sum_{\substack{l=0\\2\nmid l}}^{k-1}(ck)^{k-l}\binom{k}{l}N\sqrt{C_{l-1}C_{l+1}}(\log\log z)^{\frac{l}{2}}\right)$$

Since,

$$\frac{l^3}{\log\log z}\leq\frac{k^3}{\log\log z}$$

The third term is very negligible in comparison to the others.

$$\sum_{n\leq N}(w(n)-\log\log N)^k-\sum_{n\leq N}\left(\sum_{p\leq z}f_p(n)\right)^k=\mathcal{O}\left(\sum_{\substack{l=0\\2|l}}^{k-1}(ck)^{k-l}\binom{k}{l}C_lN(\log\log z)^{\frac{l}{2}}\right)$$
$$+\mathcal{O}\left(\sum_{\substack{l=0\\2\nmid l}}^{k-1}(ck)^{k-l}\binom{k}{l}N\sqrt{C_{l-1}C_{l+1}}(\log\log z)^{\frac{l}{2}}\right)$$

So by adjusting we get,

$$\sum_{n\le N}(w(n)-\log\log N)^k-\sum_{n\le N}\left(\sum_{p\le z}f_p(n)\right)^k=\mathcal{O}\left(N\cdot C_k\sum_{l=0}^{k-1}(ck)^{k-l}\binom{k}{l}\frac{\max\{C_l,\sqrt{C_{l-1}\cdot C_{l+1}}\}}{C_k}(\log\log z)^{\frac{l}{2}}\right)$$

$$=\mathcal{O}\left(NC_k(\log\log z)^{\frac{k}{2}}\sum_{l=0}^{k-1}\binom{k}{l}\frac{\max\{C_l,\sqrt{C_{l-1}\cdot C_{l+1}}\}}{C_k}\left(\frac{ck}{\sqrt{\log\log z}}\right)^{k-l}\right.$$

The terms in the sum is maximum when $l=k-1$. And using the fact $k\le(\log\log z)^{\frac{1}{3}}$ we get,

$$\frac{ck}{\sqrt{\log\log z}}=\mathcal{O}\left(\frac{(\log\log z)^{\frac{1}{3}}}{\sqrt{\log\log z}}\right)=\mathcal{O}\left((\log\log z)^{-\frac{1}{6}}\right)$$

$$\sum_{n\le N}(w(n)-\log\log N)^k-\sum_{n\le N}\left(\sum_{p\le z}f_p(n)\right)^k=\mathcal{O}\left(N\cdot C_k(\log\log z)^{\frac{k}{2}-\frac{1}{6}}\right)$$

We need to now replace $\log\log z$ by $\log\log N$ to complete the claim. Using the fact $z=N^{\frac{1}{k}}$ we get,

$$\log\log z=\log\log N-\log k$$

Thus we can say that $\log\log z=\mathcal{O}(\log\log N)$ Hence by replacing we get,

$$\sum_{n\le N}(w(n)-\log\log N)^k=\sum_{n\le N}\left(\sum_{p\le z}f_p(n)\right)^k+\mathcal{O}\left(N\cdot C_k(\log\log N)^{\frac{k}{2}-\frac{1}{6}}\right)$$

So by using the proposition of the first part and the last equation we can say that for k even,

$$\begin{aligned}\sum_{n\leq N}(w(n)-\log\log N)^k &= NC_k(\log\log N)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\frac{k^3}{\log\log N}\right)\right)\\ &\quad+\mathcal{O}(NC_k(\log\log N)^{\frac{k}{2}-\frac{1}{6}})\\ &= NC_k(\log\log N)^{\frac{k}{2}}\left(1+\mathcal{O}\left(\frac{k^3}{\log\log N}\right)\right)\end{aligned}$$

$$\implies \frac{1}{N}\sum_{n\leq N}\left(\frac{w(n)-\log\log N}{\sqrt{\log\log N}}\right)^k = C_k\left(1+\mathcal{O}\left(\frac{k^3}{\log\log N}\right)\right)$$

Thus we get,

$$\lim_{N\to\infty} E(X_N^k) = \lim_{N\to\infty} \frac{1}{N} \sum_{n\leq N} \left(\frac{w(n) - \log\log N}{\sqrt{\log\log N}} \right)^k = C_k$$

And for odd k similarly we have,

$$E(X_N^k) = \frac{1}{N} \sum_{n\leq N} \left(\frac{w(n) - \log\log N}{\sqrt{\log\log N}} \right)^k << C_k \frac{k^{\frac{3}{2}}}{\sqrt{\log\log N}} + \mathcal{O}(\log\log N)^{-\frac{1}{6}}$$

Thus,

$$\lim_{N\to\infty} E(X_N^k) = \lim_{N\to\infty} \frac{1}{N} \sum_{n\leq N} \left(\frac{w(n) - \log\log N}{\sqrt{\log\log N}} \right)^k = 0$$

Hence from the (corollary 1.1) we can conclude that

$$\lim_{N\to\infty} X_N \sim N(0,1)$$

Thus for any real $a \leq b$ we can say,

$$\lim_{N\to\infty} \mathbb{P}(a \leq X_N \leq b) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{x^2}{2}} dx.$$